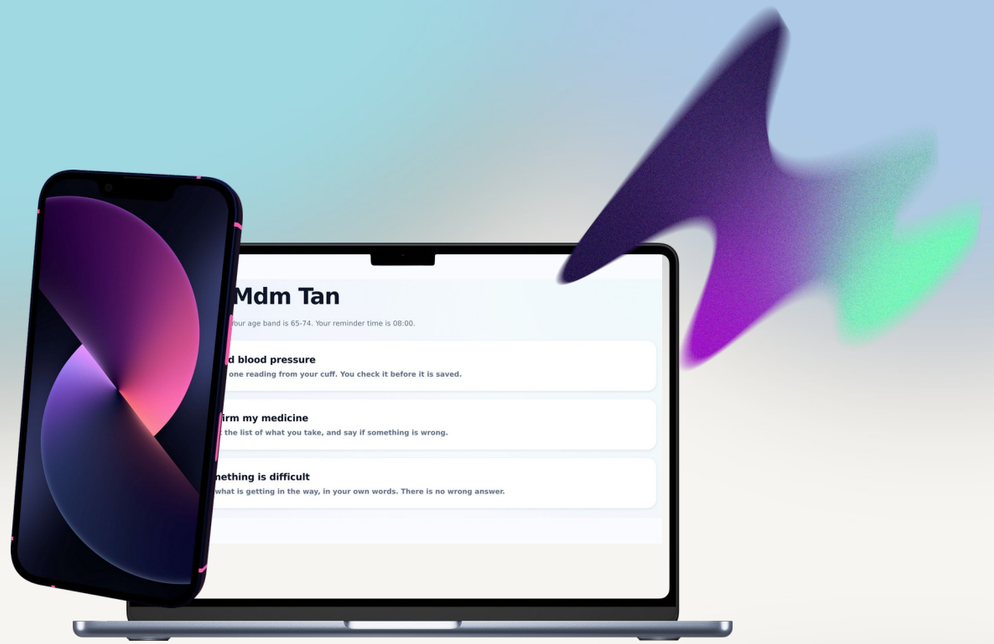
TENCENT CLOUD AI SINGAPORE HACKATHON 2026

CareSteps

Architecture & Trust Boundaries

A safety-first self-care companion for older adults managing blood pressure.

Local application • Optional model • Human confirmation



3
ordered safety gates

10
model-facing tools

13
SQLite tables

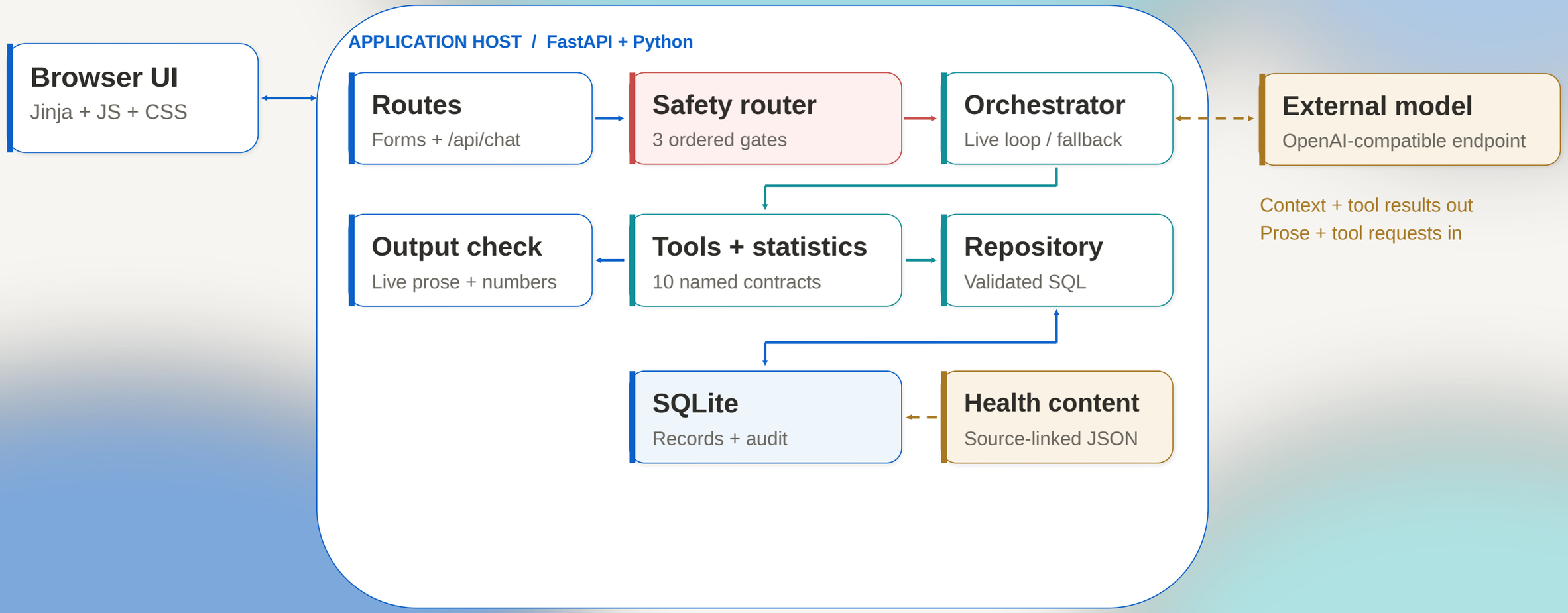
One local source of truth

Routes, safety, tools and SQLite stay together.

Healthcare Challenge 2 | AI Healthier Every Day | Competition prototype

System Architecture

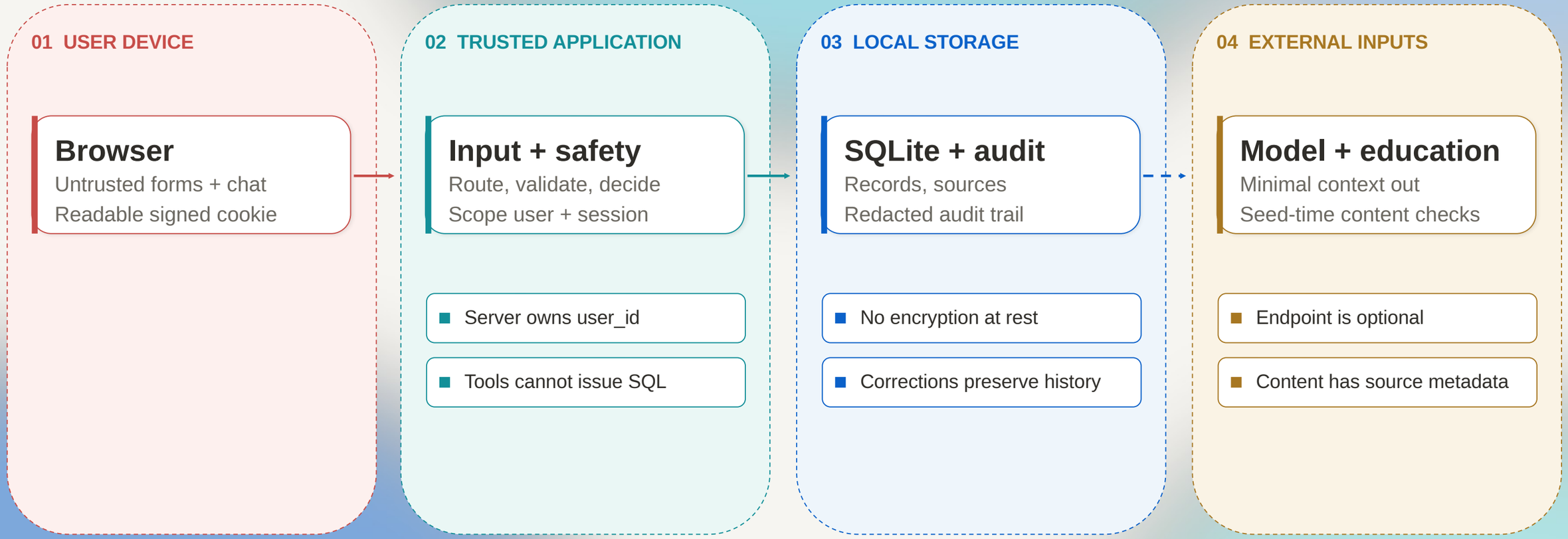
A local FastAPI application owns decisions, validation and storage; the model is optional.



Safety can return early • Forms use server validation • Model availability never owns storage

Trust Boundaries

Every model request is re-validated; browser, model and imported content have limited authority.



Prototype boundary: shared demo profile, local HTTP and readable cookie state still need production hardening.

Safety Gates Run Before the Model

The first decisive verdict ends the turn; model instructions cannot bypass this ordering.

FIXED ORDER / ONE MATCH CAN END THE TURN

01 Red flags
Symptoms + reading context

match → **Emergency / urgent guidance**
Fixed response; no model call

otherwise

02 Boundaries
Dose changes or unclear medicine

match → **Refuse + redirect**
Fixed response; no model call

otherwise

03 Clarification
Missing values or intent

match → **Ask for details**
Fixed response; no model call

WHEN ALL THREE ALLOW

Continue to:

- live model tools
- rule-based degraded support

Priority is deterministic. Detection remains limited to authored rules.

Red flags > boundaries > clarification | Safety is evaluated before the orchestrator.

Tool Calling & Human Confirmation

The server owns identity, argument checks and database operations; the model only requests named tools.

READ / COMPUTE **"What was my average blood pressure this week?"**

Safety
Gates pass

→

Request
get_period_stats
{days: 7}

→

Validate
Schema + context
user_id server-owned

→

Compute
repo + stats.py
window, gaps, n

→

Answer
Facts -> prose
Live output check

PREPARE / CONFIRM **Readings and medication reports**

Prepare
Validate + stage
No health record

→

Preview
Show draft values
User sees before save

→

Confirm
POST /api/chat/confirm
UI-only handler

→

Persist
Repository saves
Duplicate protection



10 model-facing contracts | Human confirmation protects staged health records | Scoped direct writes stay explicit

Degraded Mode & Partial Failure

Core workflows remain available when the optional model is missing, unavailable or interrupted.

No live endpoint
Unset configuration
or forced degraded

↓

Rule-based reply
mode = degraded

Call fails before tools
Timeout or unreadable
provider response

↓

Rule-based reply
mode = degraded

Tools already ran
Later failure or
round budget exhausted

↓

Tool-result fallback
mode = live + check

LOCAL WORKFLOWS REMAIN AVAILABLE

- BP entry forms
- Trends + statistics
- Medicine reports
- Visit summary

A later model failure does not undo completed tool writes; the application records actual outcomes.

Data Model & Provenance

13 SQLite tables grouped by responsibility; the repository owns request-path database access.

user_profile

Profile, preferences and simulated-data marker

HEALTH RECORDS

- care_plan
- medication_item
- bp_reading
- medication_report

SUPPORT + PREPARATION

- barrier_event
- support_action
- consultation_question
- visit_summary

KNOWLEDGE + TRACE

- education_source
- education_fact
- conversation_message
- audit_event

Preserved corrections • Unique dedupe keys • Source metadata • Audit events

Patient data is fictional and simulated; source checks do not equal clinical review.

Prompting, Grounding & Traceability

The prompt sets tone and limits; deterministic code controls routing, tools, calculations and checks.

System prompt
Role, tone, limits
caresteps.system.v3

→

Minimal context
Profile + latest
12 messages

→

Bounded tool loop
Named operations
Default: 4 rounds

→

Live output check
Known numbers + phrases
Replace on failure

PROMPT POLICY SUMMARY

- Numbers: use tool results
- Education: retrieve, then cite
- Unknowns: ask or defer
- Medication changes: refuse

INSPECTABLE AT /trace

- Active contracts + provider status
- Rule version + source IDs
- Prompt version + fingerprint
- Tool outcomes + response mode

Output checks are grounding heuristics; they do not provide clinical validation.

Key Design Trade-offs & Prototype Limits

Each constraint makes the local prototype easier to inspect and leaves a clear production task.

CHOICE	WHY IT HELPS	COST / NEXT STEP
Deterministic gates first	Immediate, inspectable safety routing	Expand clinical review + rule coverage
SQLite + local browser assets	One-process, reproducible demo	Add auth, encryption and scaling
Scoped tools + confirm flow	Validated operations + visible drafts	Add atomic turn boundaries
Optional model endpoint	Core workflows survive downtime	Verify live provider in production

170 automated tests passed | 10 model tools | 13 tables | Source-linked rules and education

Simulated data | No clinical validation | No health-system integration | No accessibility conformance audit

Handbook p. 35 | docs/decisions.md | Source assertions checked against current app/ code